Requirement Gathering:

- ## Stakeholder Analysis:
  - **Aircraft Maintenance Engineers** need to catch failures before they happen—nobody wants an engine surprise at 35,000 feet.
  - **Airline Companies** are basically trying to keep maintenance costs down while keeping planes in the air longer. Same old story: do more with less.
  - **Aerospace Engineers & Researchers** need models that actually reflect reality. They want to see how machines perform across different altitudes, temperatures, and pressures so they can tweak designs before anything gets built.
  - **Maintenance and Health Monitoring Teams** live and breathe PHM (Prognostics and Health Management). They need solid data to predict breakdowns before they occur, plus anomaly detection systems that actually work in noisy factory environments—not just in clean lab conditions.
  - **Project Supervisors** just want to know one thing: does this model match the real world, or are we fooling ourselves?
  - **Industrial Operators** need reliable abnormal sound detection. When a machine starts making weird noises, they want to know immediately, not after something expensive breaks.

- **User Stories & Use Cases:**
  - **User stories:**
  - As a Maintenance Engineer, I want to feed the system historical turbofan sensor data so I can predict Remaining Useful Life (RUL) and schedule maintenance before the engine gives out.
  - As an Airline Operator, I want dependable RUL predictions so we can plan maintenance ahead of time, cut operational costs, and—let's be honest—keep passengers safe.
  - As a Data Scientist, I want to clean up the FD004 turbofan dataset and get that multivariate sensor data and operational settings ready for an LSTM model to actually learn degradation patterns.
  - As a Factory Maintenance Engineer, I want the system to pick up abnormal sounds from fans, pumps, valves, and sliders so we can catch faults before they turn into expensive disasters.
  - As a Researcher, I want to pull MFCC features and YAMNet transfer learning embeddings from the MIMII dataset so the model gets both the low-level audio details and the bigger-picture representations.
  - As a Researcher, I want to combine MFCC and YAMNet features before training so we get better anomaly detection accuracy and a more robust model overall.
  - As a User, I want an Autoencoder that learns what "normal" sounds like and flags anomalies based on reconstruction error—ideally without me having to babysit it.

- As a System Administrator, I want one platform that handles both the FD004 turbofan data and the MIMII audio dataset. One system, two jobs.
- As a Data Scientist, I want Majority Voting across different feature representations so the final call is more reliable than any single model's guess.

User cases:

- **1. Turbofan FD004 Dataset**
  - - Upload the FD004 dataset
  - - Preprocess multivariate sensor data
  - - Deal with missing and noisy readings (because there's always noise)
  - - Normalize operational settings and sensor features
  - - Break down engine run-to-failure trajectories into usable sequences
  - - Train an LSTM on time-series data
  - - Predict Remaining Useful Life (RUL) for test trajectories
  - - Compare predictions against actual RUL
  - - Evaluate using RMSE and MAE
  - - Visualize RUL predictions and degradation trends

- **2. MIMII Dataset**
  - - Upload the MIMII machine audio dataset
  - - Pick machine type: Fan, Pump, Valve, or Slider
  - - Select four machine IDs per category
  - - Extract MFCC features from audio signals
  - - Extract YAMNet embeddings via transfer learning
  - - Combine MFCC + YAMNet into a single feature vector
  - - Train an Autoencoder on normal sound data only
  - - Reconstruct input audio representations
  - - Compute reconstruction error for anomaly detection
  - - Flag abnormal machine behavior
  - - Apply Majority Voting for final decision
  - - Evaluate performance using AUC, Accuracy, Precision, Recall, and F1-score

- **Functional Requirements:**
  - **Turbofan FD004**
    - Load FD004 training and testing trajectories
    - Handle 26-column multivariate time-series (operational settings + sensor measurements)
    - Support six different operational conditions
    - Account for two fault modes: HPC Degradation and Fan Degradation
    - Clean noisy sensor data before training
    - Normalize and format sequential data for LSTM input
    - Train LSTM model on training trajectories
    - Predict Remaining Useful Life (RUL) for test trajectories
    - Compare predicted vs true RUL values
    - Calculate RMSE, MAE, and R^2
    - Visualize prediction results and degradation trends
    - Save trained models and prediction outputs.

  - **MIMII Dataset**
    - Support MIMII dataset for industrial machine anomaly detection
    - Handle four machine types: Fan, Pump, Valve, Slider
    - Support multiple machine IDs (four per category)
    - Extract MFCC features from audio signals
    - Use YAMNet transfer learning embeddings for feature extraction
    - Combine MFCC and YAMNet features before training
    - Train Autoencoder on normal machine sounds only
    - Encoder learns compressed latent representations of normal behavior
    - Decoder reconstructs original input from latent features
    - Compute reconstruction error for anomaly detection
    - Classify small reconstruction error as normal and large as abnormal
    - Apply Majority Voting to combine prediction results
    - Evaluate using Euclidean Distance (EUC), Accuracy, Precision, Recall, and F1-score

- **Non-functional Requirements**

  Performance

  - Process large multivariate time-series datasets efficiently without making users wait
  - Generate RUL predictions within a reasonable timeframe

Security

- Restrict access to datasets, models, and predictions to authorized users only

Usability

- Keep the interface simple and straightforward for engineers and **researchers**

Reliability

- Ensure stable and consistent predictions during continuous operation

Robustness

- Maintain anomaly detection performance in noisy real-world industrial environments

Flexibility

- Support integration of multiple feature extraction methods and machine types as the system evolves

Efficiency

- Handle feature extraction and prediction on large audio datasets without performance degradation

Extensibility

- Allow easy addition of new machines and deep learning models in the future